An enzyme-discovery agent for de novo biocatalyst search

Target chemistry in → ranked, mechanism-annotated candidates out → structure-ready handoff

THE PROBLEM

Find enzymes for nylon-6,6

• synthesis: adipate + HMDA → amide

• hydrolysis: break the N66 amide

no known natural catalyst — must search

THE FRAMEWORK

target reaction contract

▼

literature + family expansion

▼

decomposed ranking

▼

mechanism + cofactor handoff

▼

structural modeling

WHAT'S NEW

Deterministic mechanism bridge:

activation mode + reactive P + cofactors

derived from the reaction, provenanced

— no LLM, fixes family-keyword errors

KEY RESULTS (run 20260914_220431, assayed panels, exhaustive audit)

1,858

candidates discovered

across 51 families

38 → 20

ranked → structure

handoff, provenanced

52 / 53

mechanism calls agree;

ddaG mis-call fixed

AUROC 0.81

step-1 NAC vs assay

(p = 0.036)

2 targets

synthesis + hydrolysis

same framework

OPEN / HONEST LIMITATIONS

- Ground-truth SfaB is unreviewed (no EC) → outside any reviewed-DB sweep; needs the literature/unannotated channel.
- Improved run spent 72% of tokens idle (a convergence/stop regression to fix).
- Geometry (angle/NAC) is a coarse negative filter, not a quantitative activity predictor.

Enzyme-discovery agent — retrieval-expansion architecture

Target chemistry drives discovery; the LLM proposes strategies, tools find entities (data-leakage firewall). Numbers from run 20260914_220431.

1 Target reaction contract

13 claims · 9 inferred

Derive chemistry from the query: direction, bond, reactive groups, activation options

2 Literature harvest

4 seed candidates

Read primary full text; seed DB expansion with tested enzymes before search

3 Mechanism-family hypotheses

6 families (3 high-promise)

Propose falsifiable mechanism folds to cover (with disqualifying observations)

4 Family survey + expansion

51 families · 1,858 discovered

Map InterPro folds; promote substrate-fit members; literature-guided expansion

5 Candidate evaluation

85 evaluated · 77 rejected

Per-candidate: EC/bond gate, substrate-fit axes, mechanism, cofactor spec, provenance

6 Decomposed ranking

38 ranked · 3 strong leads

Lexicographic spine over substrates_hosted / shape_mismatches / weakest_fg_match

7 Structural-modeling handoff

20 candidates

Per-candidate: family + mechanism (reactive P) + cofactor spec → co-fold

cross-cutting design

Data-leakage firewall

Answer enzymes never enter prompts/tools; only generic mechanism vocabulary is allowed.

Deterministic chemistry

Activation mode & cofactors derived from reaction products — no LLM, fully provenanced.

Decomposed, no composites

Substrate fit reasoned as functional-group facts, not fused Tanimoto scores.

Structural handoff

Explicit, reviewable spec: family + mechanism + cofactors travel to co-folding.

Discovery-agent → structural-modeling handoff

Per-candidate record in structural_modeling_handoff.json — three blocks are deterministically derived by the agent (green)

candidate record

identity

accession	*E2JA32*
protein_name	*... diaminopropanoate ligase*
organism, gene_names, ec_numbers	

family — NEW

interpro_family	*Dapdiamide ligase*
interpro_family_id	*IPR048223*

evidence (existing)

- structure_sources
- native_substrate_evidence
- reaction_chemistry
- structural_homology / orthologs

mechanism_evidence.activation — NEW

activation_mode	*acyl_adenylate*
reactive_phosphate	*Palpha*
nucleophile	*carboxylate O*
confidence	*high*
source / derived_from	*reaction products*

modeling_input — NEW

cofactors[] {name,ccd,count,role,basis}	*ATP×1 + MG×2*
reactive_phosphate	*Palpha*
activation_mode	*acyl_adenylate*

how the green blocks are derived (worked example: ddaG)

1 · candidate's own annotations (UniProt)

native reaction:

...+ ATP = ...+ AMP + diphosphate

cofactor comment: (none for ddaG)

↓ deterministic (no LLM)

2 · chemistry_logic derivations

derive_activation_mechanism()

AMP + diphosphate → α-phosphorus

derive_modeling_cofactors()

ATP×1; metal MG×2 (mech. default)

↓

3 · written to handoff (with provenance)

reactive_phosphate: Palpha

activation_mode: acyl_adenylate

cofactors: ATP×1, MG×2

basis: reaction products / uniprot

confidence: high

Why it matters

Family name 'Dapdiamide ligase' gives no hint of adenylation. The reaction pins it to Pα — fixing the old family-keyword mis-call that scored ddaG on Pγ.

existing field agent-derived (deterministic)

Mechanism bridge — activation chemistry from the reaction, not the family name

Deterministic classifier reads the native reaction's products; fixes the family-keyword mis-calls.

products contain NMP + diphosphate

e.g. ...+ ATP = ...+ AMP + diphosphate

→ acyl-adenylate

→ attack ATP Pα (O3A leaving axis)

products contain NDP + phosphate

e.g. ...+ ATP = ...+ ADP + phosphate

→ acyl-phosphate

→ attack ATP Pγ (O3B leaving axis)

Validation against the structural pipeline's prior (family-keyword) assignment

52 / 53

agree where reaction data exists

1

corrected (ddaG: family said none; reaction → Pα)

9

no reaction annotation → honest 'unresolved' (was silently defaulted)

ddaG (E2JA32) — the case that proves the point

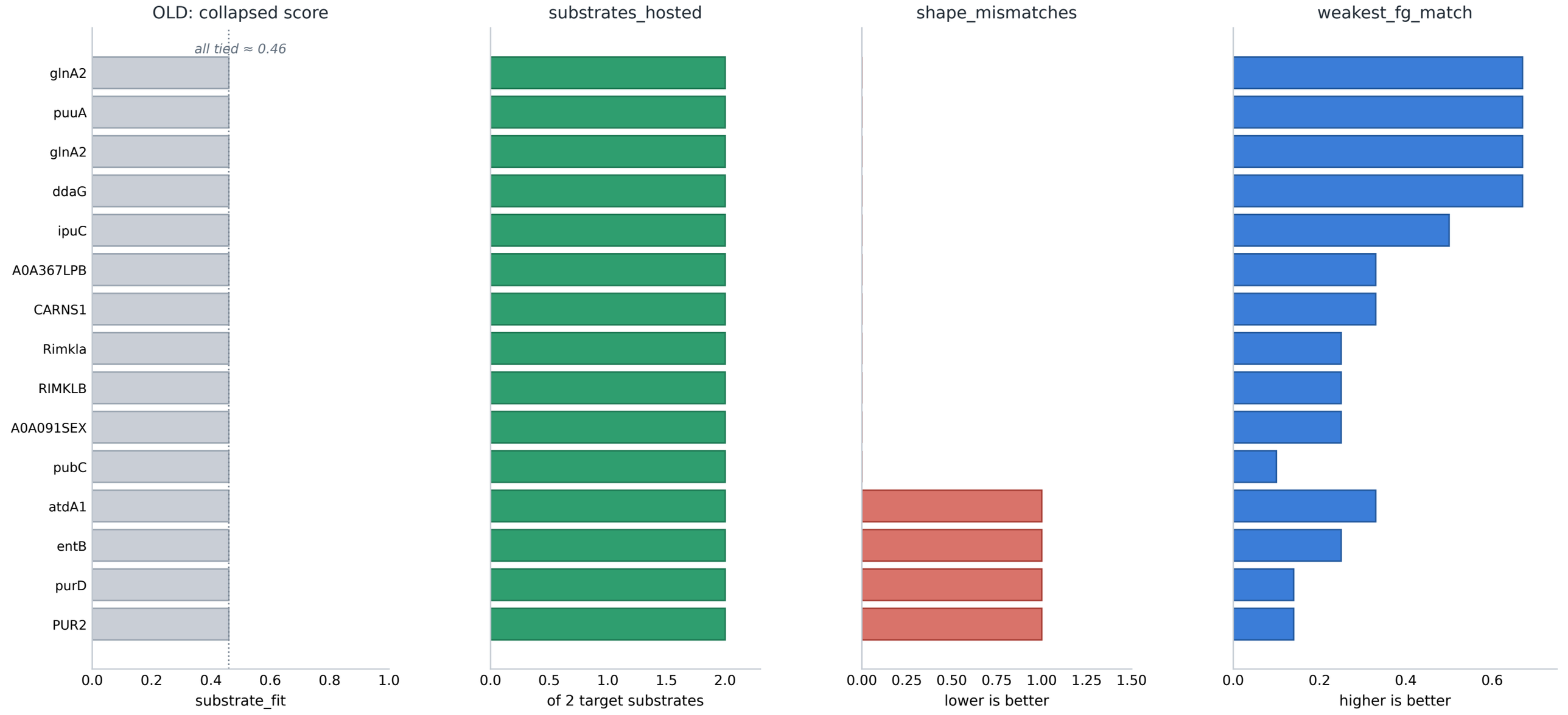
InterPro family “Fumarate–diaminopropanoate ligase” carries no adenylation hint.

Its reaction (... = ... + AMP + diphosphate) unambiguously → Pα. The old keyword map scored it on Pγ

(0 productive poses → 'inconclusive'); the reaction-derived call fixes it. Now emitted with full provenance.

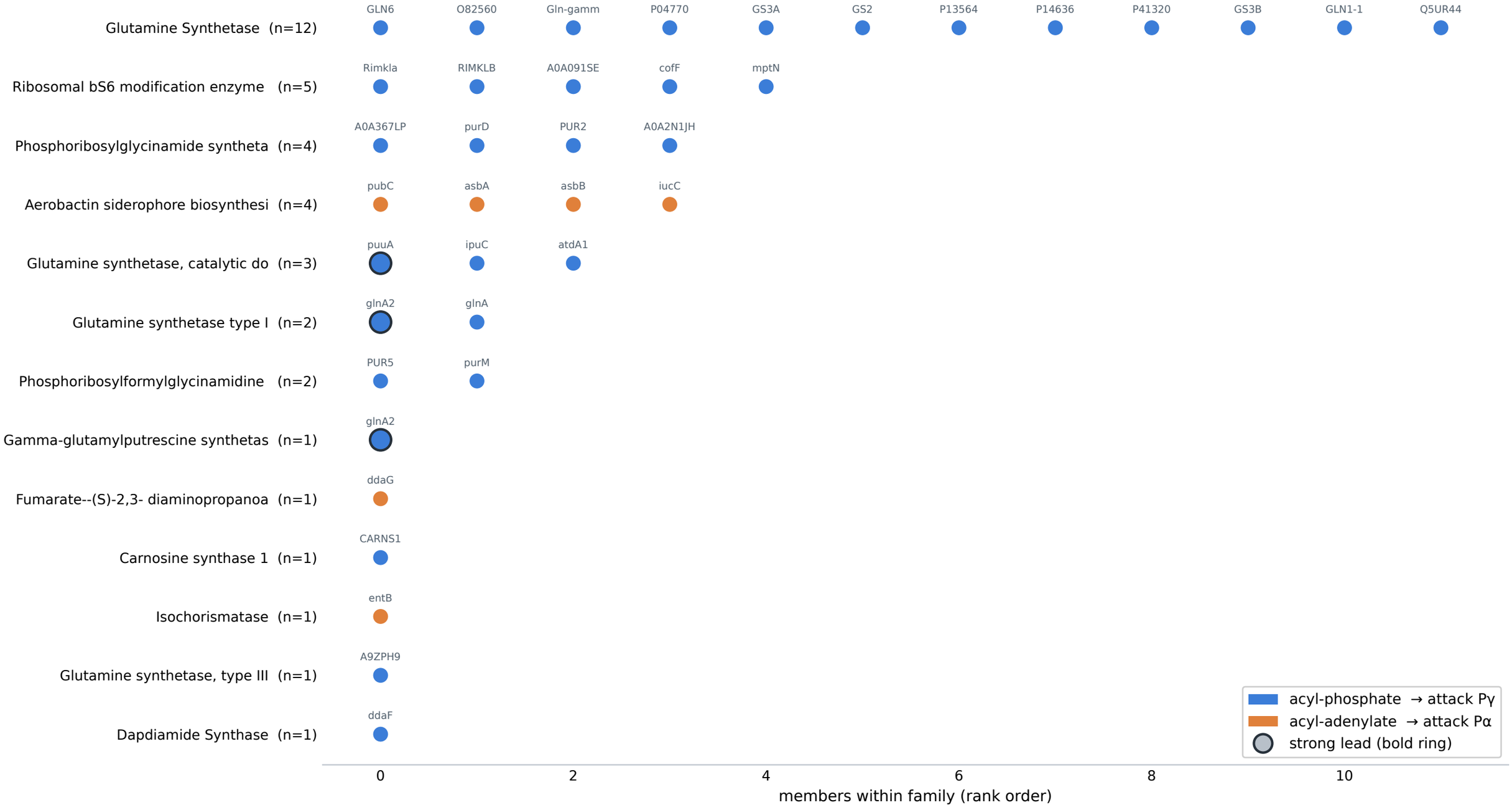
Decomposed ranking — one collapsed score replaced by three interpretable axes

Improved run, top 15. The old pipeline tied nearly every candidate at substrate_fit ≈ 0.46; the axes break the tie and expose one-sided fits.



Ranked candidate landscape — by InterPro family, colored by activation mechanism

Improved run: 38 ranked candidates across 13 families. Mechanism (reactive phosphate) derived deterministically from each enzyme's reaction.



Discovery completeness — exhaustive InterPro / EC 6.3.- sweep

Out-of-tree audit: every reviewed ATP-dependent ligase, filtered by target chemistry, vs what the agent surfaced.

12,456
reviewed EC 6.3.- ligases (Swiss-Prot universe)

↓

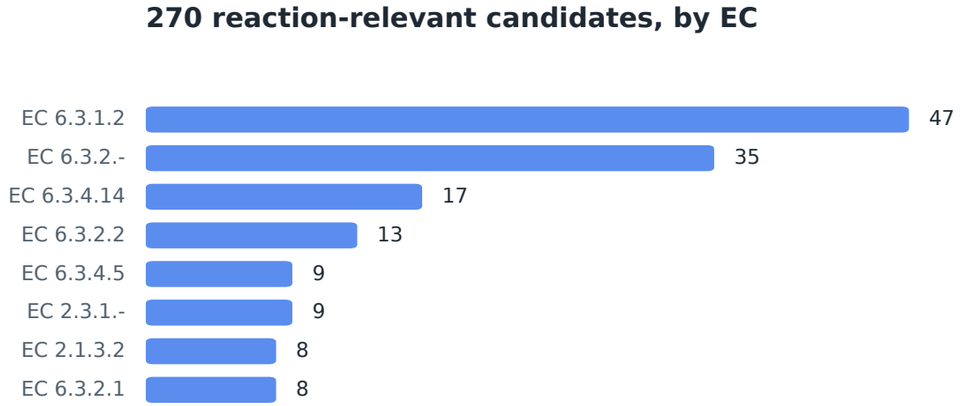
270
pass reaction-anchored chemistry filter (free diacid + amine ligation)

↓

51 / 1,858
families surveyed / candidates discovered by the agent

↓

38 → 20
ranked → handed to structural modeling

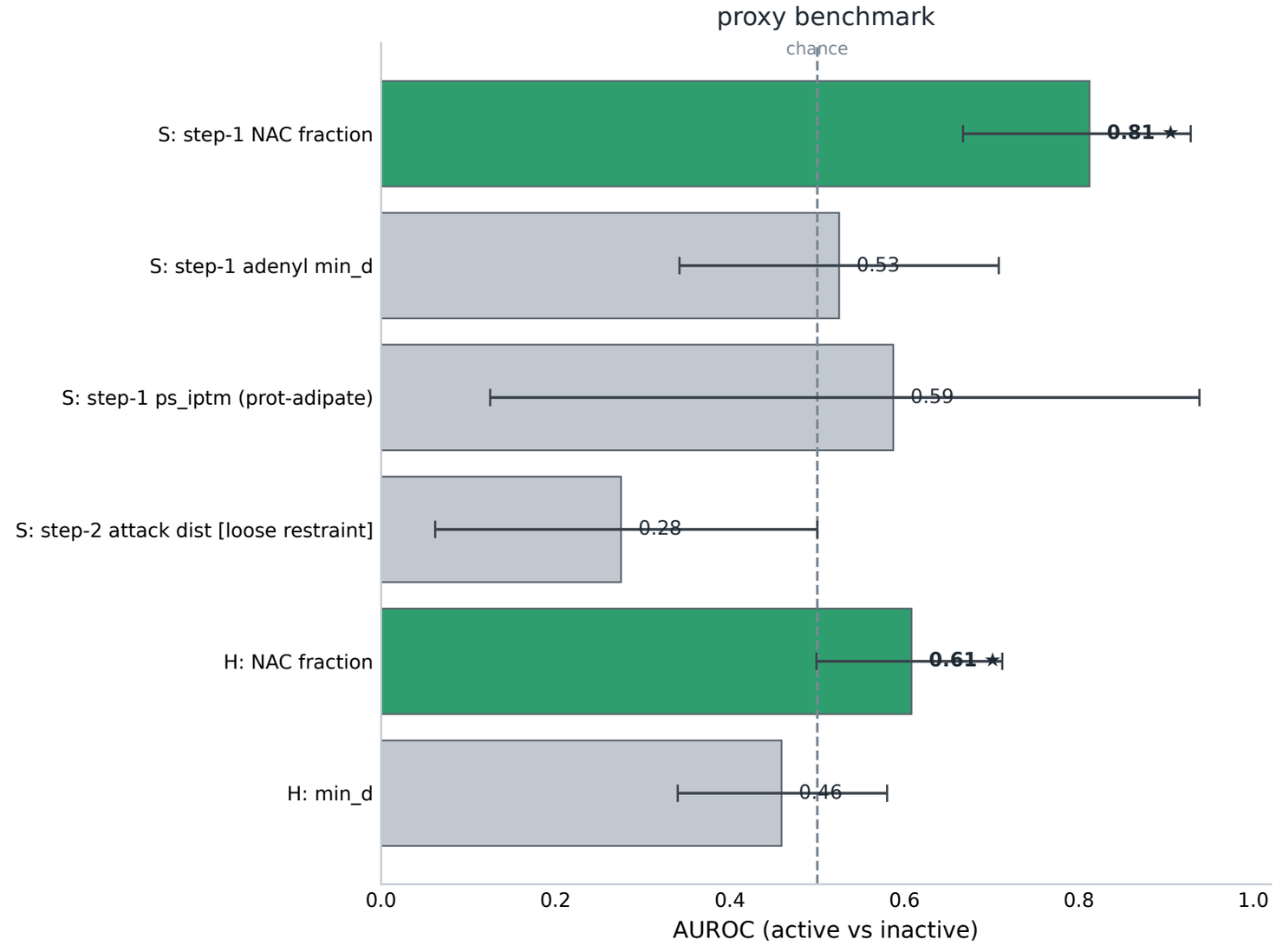


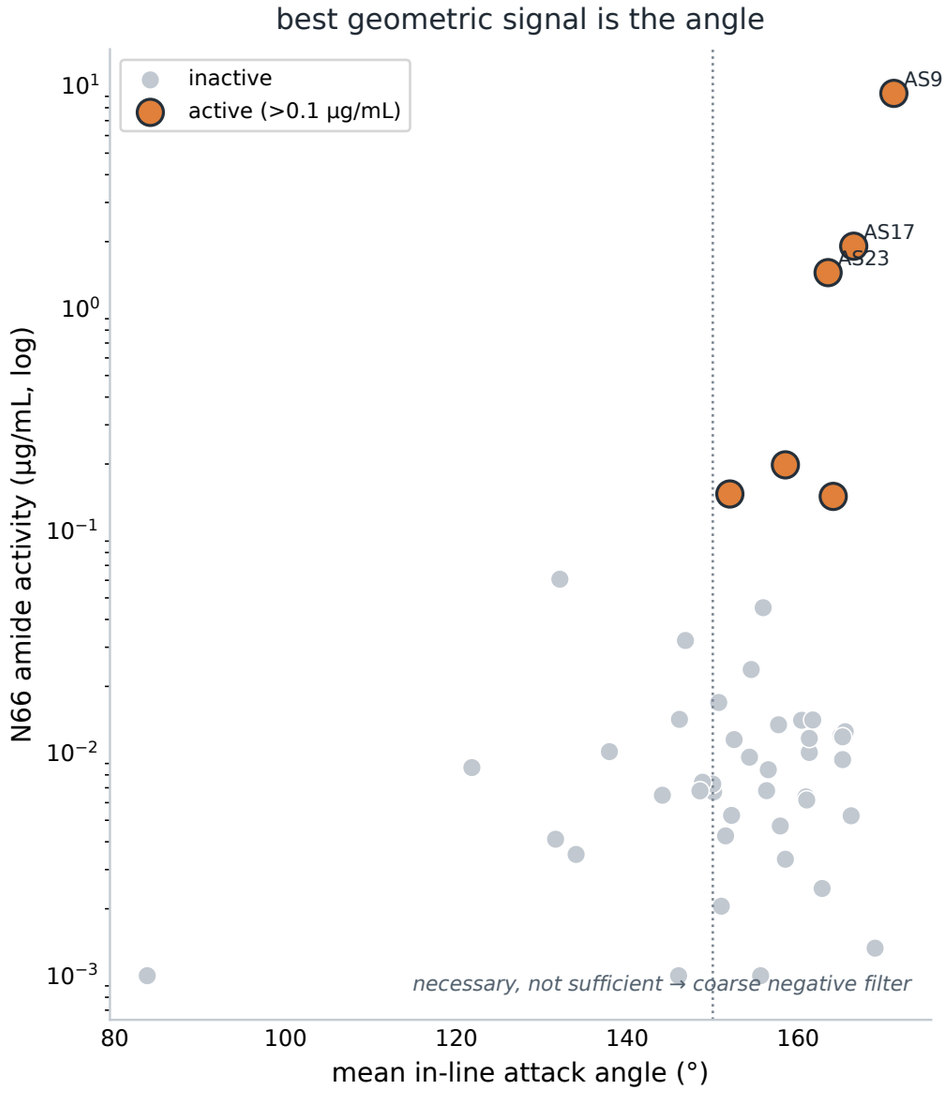
Recall ceiling (stated honestly)

The agent covers the reviewed universe — but the amide-synthetase ground truth SfaB (A0A2H4T912) is UNREVIEWED (TrEMBL, no EC), so it lies OUTSIDE this entire sweep. No reviewed-DB method recovers it; reaching it needs the literature/unannotated channel.

Geometry proxies vs measured activity — which co-fold signal discriminates actives

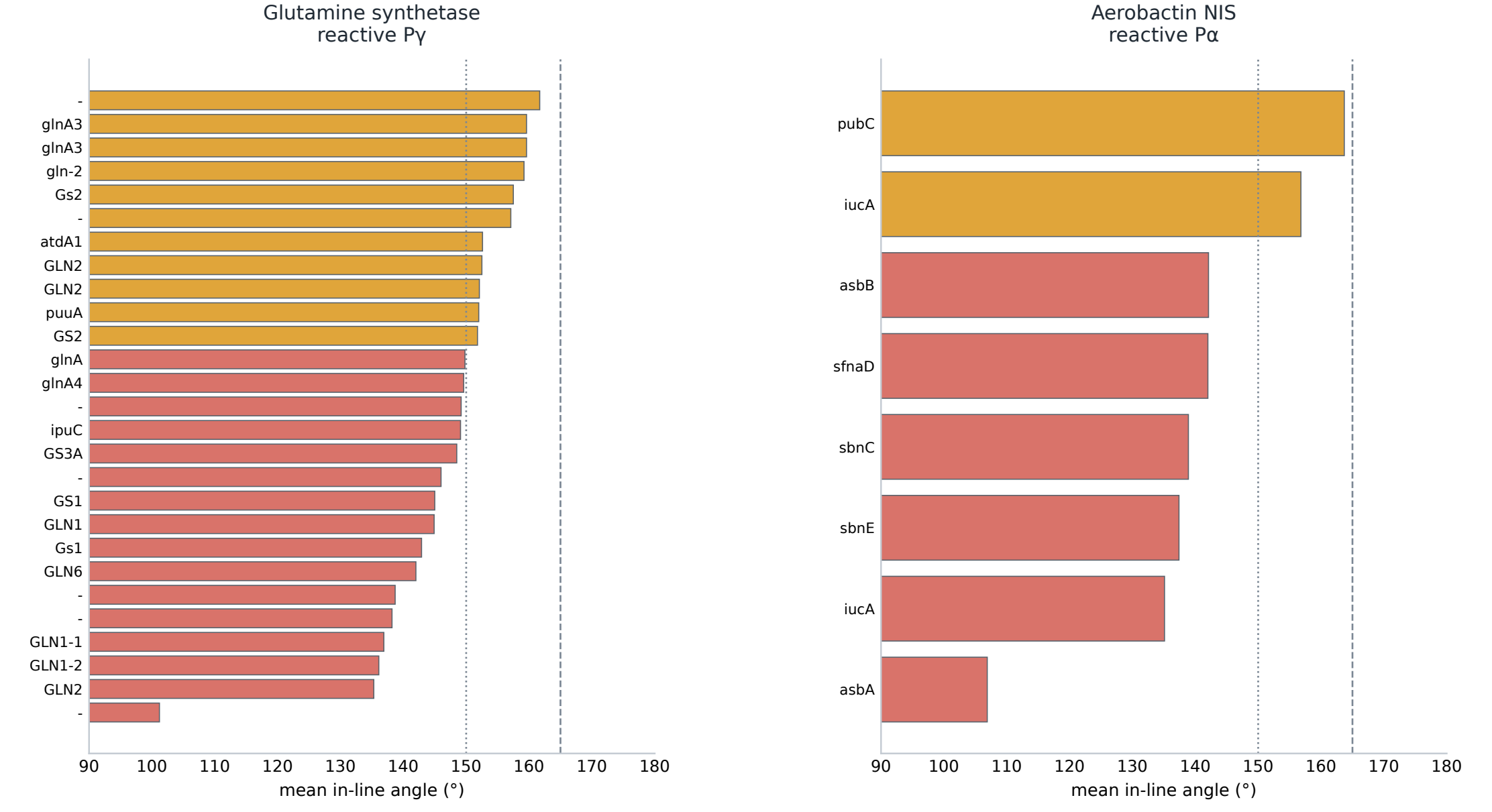
Benchmarked on assayed panels (synthetase n=49, 3 active · hydrolase n=96, 38 active). AUROC with bootstrap 95% CI; ★ = permutation p<0.05.





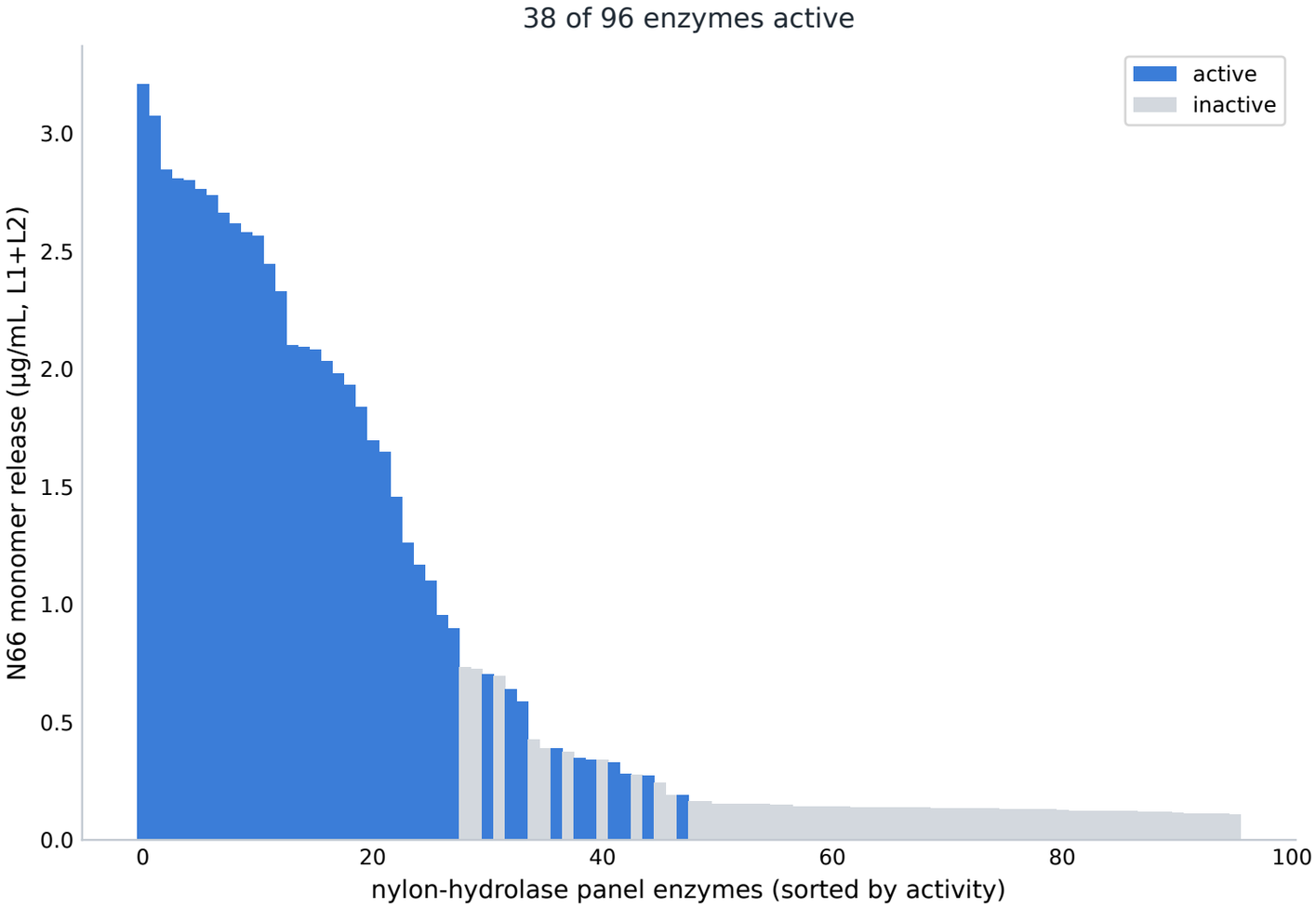
Mechanism-aware step-1 geometry — in-line attack angle, scored vs the correct ATP phosphorus

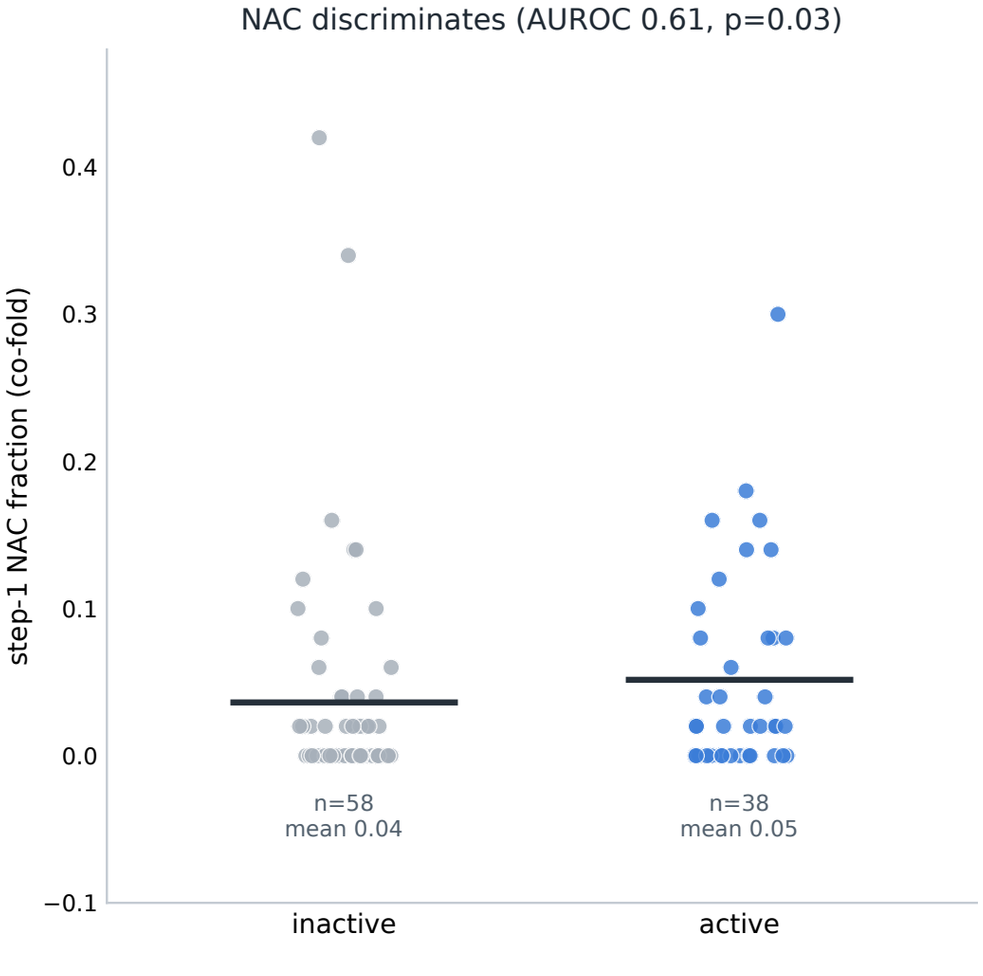
50 unrestrained Boltz-2 co-folds per enzyme; angle compared only within a mechanistic family. ≥165° strong · 150–165° compatible · <150° weak.



Same framework, second target — nylon-6,6 amide HYDROLYSIS (cleavage direction)

96-enzyme nylon-hydrolase panel (Drufva 2025), assayed for N66 monomer release. The agent's direction gate and geometry proxy transfer to the opposite reaction.





Old vs improved run — same target, cleaner ranking, but a cost regression to fix

